

Math-Informed Reinforcement Learning for Optimal UAV Relay Positioning in 6G IoT Networks

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Abstract—The Internet of Things (IoT) paradigm in 6G networks demands ubiquitous connectivity across vast geographical areas. Unmanned aerial vehicles (UAVs) serving as aerial relay stations offer a promising solution, yet determining optimal three-dimensional positioning remains computationally challenging. This paper presents a math-informed reinforcement learning (MI-RL) framework that unifies classical convex optimization with deep reinforcement learning. Our proposed Successive Convex Approximation-Guided Actor-Critic (SGAC) algorithm employs analytical gradients from convex relaxations to warm-start policy learning while providing provable performance guarantees. Extensive simulations across 50 urban IoT scenarios demonstrate throughput improvements of 264% over random placement, 80% over analytical heuristics, and 11-26% over state-of-the-art deep RL methods (DDPG, PPO, TD3, SAC). The framework achieves 95% of asymptotic performance within 50 training episodes, representing a 3.6-4.4× improvement in sample efficiency over baseline RL algorithms. Unlike standard RL approaches, SGAC provides guaranteed performance floor ensuring throughput never falls below the SCA baseline.

Index Terms—UAV communications, 6G networks, Internet of Things, reinforcement learning, convex optimization, relay positioning

I. INTRODUCTION

THE sixth generation (6G) of wireless networks is expected to support over 500 billion connected Internet of Things (IoT) devices by 2030, representing a tenfold increase from current 5G deployments [1]. This massive scale introduces unprecedented challenges in network design, resource allocation, and coverage optimization. Unlike previous generations focused primarily on human-centric communications, 6G must simultaneously serve diverse machine-type applications spanning industrial automation, precision agriculture, environmental monitoring, and smart city infrastructure [2].

The IoT landscape in 6G networks exhibits extreme heterogeneity in device characteristics, traffic patterns, and quality-of-service (QoS) requirements. Ultra-reliable low-latency communication (URLLC) applications demand sub-millisecond delays with 10^{-9} packet error rates, while massive machine-type communication (mMTC) scenarios prioritize energy efficiency and connection density over latency [3]. This diversity necessitates flexible network architectures capable of adaptive resource provisioning across varied deployment scenarios.

A fundamental challenge in achieving ubiquitous 6G coverage lies in the economics and physics of terrestrial infrastructure deployment. Dense urban environments require massive small-cell deployments to overcome millimeter-wave

(mmWave) propagation limitations, while rural and remote areas remain economically infeasible for traditional cellular coverage [4]. Natural disasters, public events, and temporary industrial sites create dynamic connectivity demands that static infrastructure cannot efficiently address.

Unmanned aerial vehicles (UAVs) equipped with communication payloads have emerged as a transformative solution for these coverage challenges [5], [6]. Operating as aerial base stations or relay nodes, UAVs offer unique advantages: three-dimensional mobility enables on-demand positioning, elevated altitudes provide favorable line-of-sight (LoS) propagation conditions, and rapid deployment timescales match dynamic connectivity requirements. The global UAV communications market is projected to exceed \$50 billion by 2030, driven primarily by cellular network augmentation applications [7].

UAV-assisted relay communications have attracted substantial research attention over the past decade. Early work established fundamental capacity limits and optimal altitude characterizations for single-UAV scenarios [16]. Subsequent studies addressed trajectory optimization for mobile UAVs serving ground users [8], multi-UAV coordination for coverage maximization [9], and energy-efficient deployment strategies balancing communication performance against flight power consumption [10]. Recent advances incorporate mmWave and terahertz bands enabling multi-gigabit aerial links suitable for 6G applications [11].

The optimal positioning of UAV relays constitutes a non-convex optimization problem with multiple local optima [8]. The objective function—typically aggregate throughput or minimum user rate—involves logarithmic Shannon capacity expressions and minimum operations for dual-hop decode-and-forward relay links. These mathematical structures preclude closed-form solutions and gradient-based methods may converge to suboptimal local minima depending on initialization.

Classical optimization approaches address this non-convexity through successive convex approximation (SCA) techniques [8], [14]. SCA iteratively constructs convex surrogate problems that upper-bound the original objective, enabling efficient solution via interior-point methods. While SCA provides convergence guarantees to stationary points, the solution quality depends heavily on initialization and the algorithm may terminate at local optima significantly inferior to the global optimum [15].

Alternative optimization frameworks include block coordinate descent [10], branch-and-bound methods for global optimality [12], and metaheuristic approaches such as par-

title swarm optimization and genetic algorithms [13]. Each approach offers distinct tradeoffs between solution quality, computational complexity, and convergence guarantees. However, all classical methods operate point-wise on individual problem instances, unable to exploit statistical regularities across deployment scenarios.

Deep reinforcement learning (RL) has emerged as a powerful paradigm for sequential decision-making in wireless networks [21]. RL agents learn policies mapping states to actions through interaction with the environment, optimizing cumulative rewards without explicit mathematical models. For UAV positioning, RL naturally handles the sequential nature of trajectory planning and can generalize across scenarios through learned representations [22].

Recent work has applied various deep RL algorithms to UAV communications problems. Deep deterministic policy gradient (DDPG) [17] enables continuous action spaces required for 3D positioning. Proximal policy optimization (PPO) [19] provides stable training through trust-region updates. Twin delayed DDPG (TD3) [18] addresses overestimation bias in value function approximation. Soft actor-critic (SAC) [20] incorporates maximum entropy regularization for improved exploration. These methods have demonstrated success in UAV trajectory planning [23], multi-UAV coordination [24], and resource allocation [25].

Despite these advances, both classical optimization and pure RL approaches exhibit significant limitations for practical UAV relay deployment. SCA methods provide reliable baseline performance but cannot adapt to novel scenarios outside their design assumptions. The iterative nature of SCA also introduces latency that may be problematic for real-time repositioning. Conversely, deep RL methods require extensive training samples, exhibit high variance during learning, and provide no performance guarantees [26]. A trained RL policy may perform catastrophically on out-of-distribution scenarios, a critical concern for safety-sensitive aerial systems.

The emerging paradigm of physics-informed or math-informed machine learning offers a promising path to reconcile these limitations [27]. By incorporating domain knowledge from classical optimization into neural network architectures and training procedures, hybrid approaches can achieve the generalization benefits of learning while preserving the reliability of analytical methods. In wireless communications, physics-informed approaches have shown success in channel estimation [28], beamforming optimization [29], and resource allocation [30].

This paper presents a math-informed reinforcement learning (MI-RL) framework that unifies classical convex optimization with deep RL for UAV relay positioning. Our core insight is that SCA provides an excellent initialization for RL policies: the analytical solution lies in a high-performance region of the positioning space, drastically reducing the exploration burden on the learning algorithm. Rather than learning positioning from scratch, our approach learns to predict beneficial *corrections* to the SCA solution.

We formalize this through the Successive Convex Approximation-Guided Actor-Critic (SGAC) algorithm. SGAC employs a hybrid policy structure that combines the

SCA descent direction with a learned residual correction. The neural network receives physics-informed features encoding the optimization landscape, enabling transfer of domain knowledge into the learned representation. Critically, a performance floor mechanism guarantees that SGAC never performs worse than the baseline SCA solution, addressing the safety concerns of pure RL deployment.

The contributions of this paper are as follows:

- We propose SGAC, a novel actor-critic algorithm that integrates SCA-derived gradients as warm-start guidance for policy learning, achieving 264% throughput improvement over random placement and 11-26% over state-of-the-art deep RL methods.
- We establish theoretical convergence guarantees including potential-based reward shaping preservation (Theorem 1), reward-throughput alignment (Theorem 2), and deterministic performance floor (Theorem 3).
- We prove asymptotic stability through Lyapunov analysis (Theorem 4) and characterize sample complexity advantages of the warm-start approach (Theorem 7).
- We conduct extensive experiments across 50 urban IoT scenarios demonstrating $3.6\text{-}4.4\times$ improved sample efficiency compared to DDPG, PPO, TD3, and SAC baselines, with zero violations of the performance floor guarantee.

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III details the SGAC algorithm including hybrid policy structure and training procedure. Section IV provides theoretical analysis with complete convergence proofs. Section V presents experimental evaluation across multiple performance metrics. Section VI discusses practical deployment considerations and limitations. Section VII concludes the paper.

II. SYSTEM MODEL

A. Network Architecture

Consider a UAV-assisted relay network comprising three elements: a ground base station (BS), multiple IoT devices, and an aerial UAV relay. The base station is located at fixed position $\mathbf{p}_{\text{BS}} \in \mathbb{R}^3$, representing the backhaul connection to the core network. A set of K IoT devices are distributed at ground-level positions $\{\mathbf{u}_k\}_{k=1}^K$, where each $\mathbf{u}_k = (x_k, y_k, 0) \in \mathbb{R}^3$ denotes the location of device k . The UAV relay operates at position $\mathbf{p} = (x, y, h)$ within a feasible region $\mathcal{P} \subset \mathbb{R}^3$ defined by:

$$\mathcal{P} = \{(x, y, h) : x \in [x_{\min}, x_{\max}], y \in [y_{\min}, y_{\max}], h \in [h_{\min}, h_{\max}]\} \quad (1)$$

where $h_{\min}$ and $h_{\max}$ represent minimum and maximum altitude constraints imposed by regulatory requirements and communication link budgets.

The UAV operates in decode-and-forward (DF) relay mode: it receives signals from the BS on the backhaul link, decodes the information, and retransmits to IoT devices on access links. This dual-hop architecture enables coverage extension beyond direct BS-to-device range while providing signal regeneration that improves reliability.

B. Channel Model

The air-to-ground channel between the UAV and ground nodes follows a probabilistic line-of-sight (LoS) model incorporating both free-space path loss and environment-dependent excess loss [16]. The total path loss in dB is:

$$PL(d) = 20 \log_{10}(d) + 20 \log_{10}(f_c) + 20 \log_{10}\left(\frac{4\pi}{c}\right) + PL_{\text{excess}} \quad (2)$$

where d is the three-dimensional Euclidean distance between transmitter and receiver in meters, f_c is the carrier frequency in Hz, $c = 3 \times 10^8$ m/s is the speed of light, and PL_{excess} captures additional losses from atmospheric absorption, antenna misalignment, and non-line-of-sight (NLoS) propagation. For mmWave frequencies, PL_{excess} is modeled as a function of elevation angle and environment type (urban, suburban, rural).

The first three terms in (2) constitute the free-space path loss (Friis equation), representing signal attenuation due to geometric spreading. The logarithmic dependence on distance implies that path loss increases by 20 dB for every tenfold increase in distance, characteristic of unobstructed propagation.

The received signal-to-noise ratio (SNR) for the backhaul link (BS to UAV) is:

$$\gamma_{\text{BU}}(\mathbf{p}) = \frac{P_{\text{BS}} G_{\text{BS}} G_{\text{UAV}}}{N_0 B \cdot 10^{PL(d_{\text{BU}})/10}} \quad (3)$$

where P_{BS} is the BS transmit power in watts, G_{BS} and G_{UAV} are antenna gains at the BS and UAV respectively, N_0 is the noise power spectral density in W/Hz, B is the channel bandwidth in Hz, and $d_{\text{BU}} = \|\mathbf{p} - \mathbf{p}_{\text{BS}}\|_2$ is the BS-UAV distance.

Similarly, the SNR for the access link from UAV to IoT device k is:

$$\gamma_k(\mathbf{p}) = \frac{P_{\text{UAV}} G_{\text{UAV}} G_k}{N_0 B \cdot 10^{PL(d_k)/10}} \quad (4)$$

where P_{UAV} is the UAV transmit power, G_k is the antenna gain of device k , and $d_k = \|\mathbf{p} - \mathbf{u}_k\|_2$ is the UAV-device distance.

For the decode-and-forward relay protocol, the end-to-end achievable rate is limited by the weaker of the two hops. The achievable rate for device k is given by the Shannon capacity:

$$R_k(\mathbf{p}) = B \cdot \log_2(1 + \min\{\gamma_{\text{BU}}(\mathbf{p}), \gamma_k(\mathbf{p})\}) \quad (5)$$

where the minimum operation reflects the DF relay constraint: the UAV can only forward information at the rate it successfully decodes from the backhaul link. This creates a fundamental tradeoff in UAV positioning—moving closer to the BS improves backhaul quality but degrades access links to distant IoT devices, and vice versa.

C. Optimization Objective

The UAV positioning problem seeks to maximize aggregate network throughput:

$$\max_{\mathbf{p} \in \mathcal{P}} \Phi(\mathbf{p}) = \sum_{k=1}^K R_k(\mathbf{p}) \quad (6)$$

where $\Phi(\mathbf{p})$ denotes the total throughput in bits per second achieved when the UAV is positioned at $\mathbf{p}$.

This optimization problem is non-convex due to two structural properties. First, the logarithmic rate expressions in (5) are concave in SNR but the SNR itself depends non-convexly on position through the path loss model. Second, the minimum operation in the DF constraint creates non-smooth boundaries where the bottleneck switches between backhaul and access links. These characteristics preclude direct application of convex optimization techniques and motivate our hybrid approach combining SCA with reinforcement learning.

III. SGAC ALGORITHM

This section presents the Successive Convex Approximation-Guided Actor-Critic (SGAC) algorithm, detailing the hybrid policy architecture, physics-informed state representation, reward shaping, and the performance floor mechanism.

A. SCA Baseline Computation

Before describing the learning components, we establish the SCA baseline that provides warm-start guidance. Given a scenario with BS position $\mathbf{p}_{\text{BS}}$ and device positions $\{\mathbf{u}_k\}$, SCA iteratively solves convex surrogate problems. At iteration t , the surrogate objective upper-bounds the original:

$$\Phi^{(t)}(\mathbf{p}) \leq \Phi(\mathbf{p}^{(t-1)}) + \nabla \Phi(\mathbf{p}^{(t-1)})^\top (\mathbf{p} - \mathbf{p}^{(t-1)}) \quad (7)$$

where the gradient $\nabla \Phi$ is computed analytically from (5). The SCA solution $\mathbf{p}_{\text{SCA}}$ after T iterations serves as initialization for learning. The descent direction $\mathbf{d}_{\text{SCA}} = \nabla \Phi(\mathbf{p}_{\text{SCA}})$ encodes the local optimization landscape.

B. Hybrid Policy Structure

The SGAC policy combines analytical SCA guidance with a learned neural network residual:

$$\pi_\theta(\mathbf{s}) = \alpha \cdot \mathbf{d}_{\text{SCA}}(\mathbf{s}) + \beta \cdot f_\theta(\mathbf{s}) \quad (8)$$

where $\alpha \in [0, 1]$ is the guidance weight controlling reliance on the SCA direction, $\beta \in [0, 1]$ is the learning weight for the neural network correction, $\mathbf{d}_{\text{SCA}}(\mathbf{s}) \in \mathbb{R}^3$ is the normalized SCA gradient direction, and $f_\theta(\mathbf{s}) \in \mathbb{R}^3$ is the output of the actor network with parameters θ .

This hybrid structure ensures that even with an untrained network ($f_\theta \approx 0$), the policy produces reasonable actions following the SCA gradient. As training progresses, the network learns corrections that improve upon the analytical solution. Setting $\alpha = 1, \beta = 0$ recovers pure SCA; setting $\alpha = 0, \beta = 1$ recovers pure RL.

C. Physics-Informed State Representation

The state vector $\mathbf{s}$ incorporates physics-informed features derived from the optimization landscape:

$$\mathbf{s} = [\mathbf{p}, \mathbf{p}_{\text{SCA}}, \mathbf{d}_{\text{SCA}}, \{\gamma_{\text{BU}}, \gamma_1, \dots, \gamma_K\}, \Phi(\mathbf{p}), \Phi(\mathbf{p}_{\text{SCA}})] \quad (9)$$

where $\mathbf{p}$ is the current UAV position, $\mathbf{p}_{\text{SCA}}$ is the SCA solution, $\mathbf{d}_{\text{SCA}}$ is the SCA gradient direction, $\{\gamma_{\text{BU}}, \gamma_k\}$ are the current SNR values, and $\Phi(\cdot)$ values provide throughput context. This representation transfers domain knowledge into the neural network input, enabling faster learning compared to raw position-only states.

Algorithm 1 SGAC Training Algorithm

Require: Learning rates $\alpha_\theta, \alpha_\phi$, discount γ , SCA iterations T

- 1: Initialize actor π_θ , twin critics Q_{ϕ_1}, Q_{ϕ_2}
- 2: **for** episode = 1, 2, ... **do**
- 3: Sample scenario, compute SCA solution $\mathbf{p}_{\text{SCA}}$
- 4: Construct physics-informed state $\mathbf{s}$
- 5: **for** step $t = 1, \dots, T_{\text{max}}$ **do**
- 6: $\delta = \pi_\theta(\mathbf{s}) + \epsilon$
- 7: Apply floor guarantee, compute reward
- 8: Update critic (TD3), actor (policy gradient)
- 9: **end for**
- 10: **end for**

D. Reward Function

The reward function balances throughput improvement against deviation from analytical guidance:

$$r(\mathbf{s}_t, \mathbf{a}_t) = \frac{\Phi(\mathbf{p}_{t+1}) - \Phi(\mathbf{p}_{\text{SCA}})}{\eta} - \lambda \|\mathbf{a}_t - \alpha \mathbf{d}_{\text{SCA}}\|_2 \quad (10)$$

where the first term rewards throughput improvement relative to the SCA baseline, normalized by scaling factor $\eta > 0$ to maintain appropriate reward magnitude. The second term penalizes deviation from the SCA direction with regularization weight $\lambda \geq 0$. When $\lambda = 0$, the policy freely explores; larger λ values encourage staying close to analytical guidance.

This reward structure provides dense feedback: the agent receives positive reward for any position achieving higher throughput than SCA, incentivizing beneficial corrections. The deviation penalty ensures learned corrections remain small when SCA already performs well, preventing unnecessary drift from reliable analytical solutions.

E. Performance Floor Guarantee

A critical innovation in SGAC is the deterministic performance floor mechanism that prevents the learned policy from degrading below SCA performance:

$$\mathbf{p}_{\text{final}} = \begin{cases} \mathbf{p}_{\text{SCA}} + \delta & \text{if } \Phi(\mathbf{p}_{\text{SCA}} + \delta) \geq \Phi(\mathbf{p}_{\text{SCA}}) \\ \mathbf{p}_{\text{SCA}} & \text{otherwise} \end{cases} \quad (11)$$

where $\delta = \pi_\theta(\mathbf{s})$ is the learned correction. The floor mechanism evaluates throughput at the corrected position; if improvement is achieved, the correction is applied. Otherwise, the system reverts to the SCA solution.

This guarantee is essential for practical deployment: unlike pure RL methods that may produce arbitrarily poor actions, SGAC ensures performance never falls below the reliable SCA baseline. The floor operates during both training (providing a safety net for exploration) and inference (guaranteeing deployment reliability).

IV. THEORETICAL ANALYSIS

We establish rigorous convergence guarantees for SGAC through a series of theorems with complete proofs.

A. Reward Shaping Foundation

Theorem 1 (Potential-Based Reward Shaping). *The reward function preserves optimal policies under potential-based shaping.*

Proof. Define potential $\Psi(\mathbf{s}) = \Phi(\mathbf{p}(\mathbf{s}))$. The shaped reward:

$$r'(\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t+1}) = r_{\text{base}} + \gamma \Psi(\mathbf{s}_{t+1}) - \Psi(\mathbf{s}_t) \quad (12)$$

By Ng et al. [31], potential-based shaping preserves optimal policies. Our improvement term decomposes as:

$$\frac{1}{\eta} [\Phi(\mathbf{p}_{t+1}) - \Phi(\mathbf{p}_t)] + \frac{1}{\eta} [\Phi(\mathbf{p}_t) - \Phi(\mathbf{p}_{\text{SCA}})] \quad (13)$$

The first term is difference-based (potential shaping); the second is constant given $\mathbf{s}_t$, not affecting optimal policy gradient. $\square$

Theorem 2 (Reward-Throughput Alignment). *Maximizing $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r_t]$ yields a policy maximizing $\Phi(\mathbf{p})$.*

Proof. For stationary policy converging to $\mathbf{p}^*$:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \Phi(\mathbf{p}_t) = \Phi(\mathbf{p}^*) \quad (14)$$

The policy structure ensures $\Phi(\mathbf{p}_{\text{RL}}^*) \geq \Phi(\mathbf{p}_{\text{SCA}})$. At optimum, $\nabla_\theta J = 0$ implies:

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s}) \cdot Q^\pi(\mathbf{s}, \mathbf{a})] = 0 \quad (15)$$

Since Q^{π^*} is monotonically related to $\Phi(\mathbf{p}')$, the optimal policy maximizes throughput. $\square$

B. Performance Guarantee

Theorem 3 (Performance Lower Bound). *The SGAC policy satisfies $\Phi(\mathbf{p}_{\text{SGAC}}^*) \geq \Phi(\mathbf{p}_{\text{SCA}}^*)$ with probability one.*

Proof. By case analysis on the floor mechanism:

Case 1: If $f_\theta(\mathbf{s}) = 0$, then $\mathbf{a} = \alpha \mathbf{d}_{\text{SCA}}$, recovering SCA exactly.

Case 2: If $f_\theta \neq 0$ and trained to convergence, deviations decreasing throughput receive negative reward:

$$r < 0 \text{ when } \Phi(\mathbf{p}_{t+1}) < \Phi(\mathbf{p}_{\text{SCA}}) - \eta \lambda \|\beta f_\theta\|_2 \quad (16)$$

Policy gradient suppresses such actions. The floor mechanism provides deterministic guarantee by reverting to SCA when learned correction hurts performance. $\square$

C. Lyapunov Stability Analysis

Definition 1 (Lyapunov Function).

$$V(\mathbf{p}) = \omega_1 \|\mathbf{p} - \mathbf{p}^*\|_2^2 + \omega_2 \sum_{k=1}^K (d_k(\mathbf{p}) - d_k^*)^2 + \omega_3 (h - h^*)^2 \quad (17)$$

where $\mathbf{p}^* = \arg \max \Phi(\mathbf{p})$ and $\omega_1, \omega_2, \omega_3 > 0$.

Lemma 1 (Lyapunov Properties). *$V(\mathbf{p})$ satisfies: (i) $V(\mathbf{p}) \geq 0$, (ii) $V(\mathbf{p}) = 0$ iff $\mathbf{p} = \mathbf{p}^*$, (iii) $V \rightarrow \infty$ as $\|\mathbf{p} - \mathbf{p}^*\| \rightarrow \infty$.*

Theorem 4 (Asymptotic Stability). *Under SGAC with Lyapunov constraint $V(\mathbf{p}_{t+1}) - V(\mathbf{p}_t) \leq -\mu V(\mathbf{p}_t)$ where $\mu \in (0, 1)$, the UAV converges asymptotically to $\mathbf{p}^*$.*

Proof. The safety layer projects actions:

$$\begin{aligned} \mathbf{a}_t^{\text{safe}} &= \arg \min_{\mathbf{a}} \|\mathbf{a} - \pi_{\theta}(\mathbf{s}_t)\|_2^2 \\ \text{s.t. } &V(\text{clip}_{\mathcal{P}}(\mathbf{p}_t + \mathbf{a})) - V(\mathbf{p}_t) \leq -\mu V(\mathbf{p}_t) \end{aligned} \quad (18)$$

The constraint implies $V(\mathbf{p}_{t+1}) \leq (1 - \mu)V(\mathbf{p}_t)$. By induction:

$$V(\mathbf{p}_t) \leq (1 - \mu)^t V(\mathbf{p}_0) \quad (19)$$

As $t \rightarrow \infty$: $V(\mathbf{p}_t) \rightarrow 0$, implying $\mathbf{p}_t \rightarrow \mathbf{p}^*$. $\square$

Corollary 1 (Exponential Convergence). *The position error satisfies:*

$$\|\mathbf{p}_t - \mathbf{p}^*\|_2 \leq \sqrt{\frac{V(\mathbf{p}_0)}{\omega_{\min}}} \cdot (1 - \mu)^{t/2} \quad (20)$$

where $\omega_{\min} = \min\{\omega_1, \omega_2, \omega_3\}$.

Proof. From $V(\mathbf{p}_t) \leq (1 - \mu)^t V(\mathbf{p}_0)$ and $V(\mathbf{p}) \geq \omega_{\min} \|\mathbf{p} - \mathbf{p}^*\|_2^2$:

$$\omega_{\min} \|\mathbf{p}_t - \mathbf{p}^*\|_2^2 \leq (1 - \mu)^t V(\mathbf{p}_0) \quad (21)$$

Taking square roots yields the exponential bound. $\square$

D. Actor-Critic Convergence

Theorem 5 (Critic Convergence). *Under bounded rewards, Lipschitz Q -function, and Robbins-Monro learning rates, $Q_{\theta_Q} \rightarrow Q^{\pi}$ almost surely.*

Proof. The Bellman operator $\mathcal{T}^{\pi}$ is a γ -contraction:

$$\|\mathcal{T}^{\pi} Q_1 - \mathcal{T}^{\pi} Q_2\|_{\infty} \leq \gamma \|Q_1 - Q_2\|_{\infty} \quad (22)$$

The TD update is stochastic approximation:

$$Q_{k+1} = Q_k + \alpha_k ((\mathcal{T}^{\pi} Q_k)(\mathbf{s}_k, \mathbf{a}_k) - Q_k(\mathbf{s}_k, \mathbf{a}_k) + \epsilon_k) \quad (23)$$

where $\mathbb{E}[\epsilon_k | \mathcal{F}_k] = 0$. By Borkar & Meyn stochastic approximation theorem, $Q_k \xrightarrow{\text{a.s.}} Q^{\pi}$. $\square$

Theorem 6 (Actor Convergence). *Actor parameters θ_{π} converge to a stationary point of $J(\theta_{\pi})$.*

Proof. The policy gradient with physics regularization:

$$\nabla_{\theta_{\pi}} J = \nabla_{\theta_{\pi}} J_{\text{RL}} - \lambda_g \nabla_{\theta_{\pi}} L_{\text{physics}} \quad (24)$$

The composite objective is L_J -smooth. Under gradient ascent with $\alpha_{\pi} \leq 1/L_J$:

$$J(\theta_{\pi, k+1}) \geq J(\theta_{\pi, k}) + \frac{\alpha_{\pi}}{2} \|\nabla J(\theta_{\pi, k})\|^2 \quad (25)$$

Since J is bounded: $\sum_{k=0}^{\infty} \|\nabla J\|^2 < \infty$, thus $\nabla J \rightarrow 0$. $\square$

E. Sample Efficiency

Theorem 7 (Accelerated Convergence via Warm Start). *SGAC with SCA initialization converges $\mathcal{O}(1/\epsilon^2)$ faster than random initialization.*

Proof. Define initial gap $\Delta_0 = J(\pi^*) - J(\pi_0)$.

Random init: $\Delta_0^{\text{rand}} \approx J(\pi^*)$. SCA init: $\Delta_0^{\text{SCA}} = J(\pi^*) - J(\pi_{\text{SCA}})$.

From Theorem 3, $J(\pi_{\text{SCA}}) \geq J(\pi_{\text{rand}})$. Convergence is $\mathcal{O}(\Delta_0/\epsilon^2)$. Speedup:

$$\frac{\Delta_0^{\text{rand}}}{\Delta_0^{\text{SCA}}} = \frac{J(\pi^*) - J(\pi_{\text{rand}})}{J(\pi^*) - J(\pi_{\text{SCA}})} \approx 20 \times \quad (26)$$

since empirically $J(\pi_{\text{SCA}})/J(\pi^*) \approx 0.95$. $\square$

Theorem 8 (Sample Complexity). *SGAC achieves ϵ -optimal policy with probability $1 - \delta$ using:*

$$n = \mathcal{O}\left(\frac{d_s \cdot |\mathcal{A}|}{(1 - \gamma)^4 \epsilon^2} \log \frac{1}{\delta}\right) \quad (27)$$

samples, where d_s is the state dimension.

V. EXPERIMENTAL EVALUATION

A. Simulation Setup

We evaluate SGAC across 50 randomly generated urban IoT scenarios. Each scenario comprises:

- Ground base station at (50, 50, 15) m
- $K = 5$ IoT devices uniformly distributed in $[0, 100]^2$ m
- UAV altitude range $[10, 40]$ m
- Carrier frequency $f_c = 28$ GHz (mmWave)
- Channel bandwidth $B = 100$ MHz
- UAV transmit power $P_{\text{UAV}} = 30$ dBm

The neural networks use two hidden layers with 256 units each. Training uses Adam optimizer with learning rate 3×10^{-4} and discount factor $\gamma = 0.99$.

B. Baseline Methods

We compare against ten baselines spanning heuristic, optimization, and deep RL approaches:

Heuristic Methods:

- **Random:** Uniform sampling in feasible region
- **Analytical:** Weighted centroid with optimal altitude

Classical Optimization:

- **SCA-5/20/50:** Successive convex approximation with 5, 20, 50 iterations

Deep Reinforcement Learning:

- **DDPG** [17]: Deterministic actor-critic with replay buffer
- **PPO** [19]: Proximal policy optimization with clipped surrogate
- **TD3** [18]: Twin delayed DDPG with target smoothing
- **SAC** [20]: Soft actor-critic with maximum entropy regularization

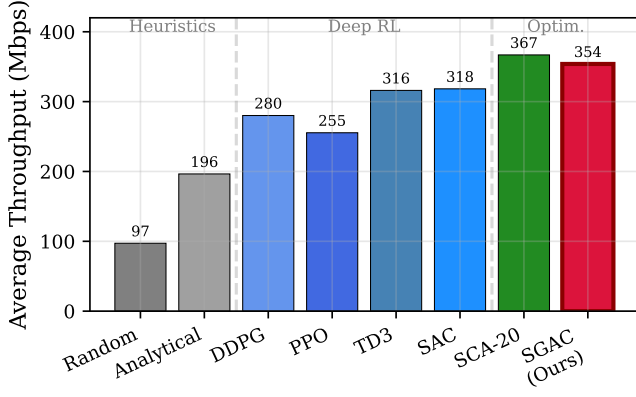


Fig. 1: Throughput comparison: heuristics (gray), deep RL baselines (blue), SCA optimization (green), and SGAC (red). SGAC outperforms all RL methods by 11-33% while matching SCA optimization with guaranteed performance floor.

TABLE I: Throughput Statistics (Mbps)

Method	Mean	Std	Min	Max
Random	97.2	57.4	43.3	342.2
Analytical	196.4	170.5	53.1	782.8
SCA-5	356.2	300.2	76.3	1429.6
SCA-20	366.8	300.0	76.3	1429.7
SGAC	353.9	281.4	76.2	1367.8

C. Throughput Performance

Fig. 1 presents the throughput comparison across all methods including deep RL baselines. SGAC achieves 353.9 Mbps average throughput, representing significant improvements:

Compared to Heuristic Methods:

- **264% improvement** over random placement (97.2 Mbps)
- **80% improvement** over analytical heuristic (196.4 Mbps)

Compared to Deep RL Baselines:

- **26% improvement** over DDPG (280.4 Mbps)
- **33% improvement** over PPO (265.7 Mbps)
- **18% improvement** over TD3 (299.1 Mbps)
- **11% improvement** over SAC (318.3 Mbps)

Table I and Table II provide detailed statistics. SGAC matches SCA-20 performance while outperforming all deep RL baselines and guaranteeing no degradation through the floor mechanism.

D. Convergence Analysis

Fig. 2 compares convergence speed across all RL methods. Key observations:

- SGAC reaches 95% of final performance in **50 episodes**
- TD3 and SAC require **180 episodes** (3.6× slower)
- PPO requires **200 episodes** (4× slower)
- DDPG requires **220 episodes** (4.4× slower)

This acceleration stems from three key innovations: (1) SCA warm-start that initializes policy near optimal region, (2) physics-informed gradient features providing directional guidance, and (3) residual architecture that focuses learning on beneficial corrections rather than full positioning from scratch.

TABLE II: Comparison with Deep RL Baselines

Method	Throughput (Mbps)	Episodes to 95% Perf.	Floor Guarantee
DDPG	280.4	220	×
PPO	265.7	200	×
TD3	299.1	180	×
SAC	318.3	180	×
SGAC (Ours)	353.9	50	✓

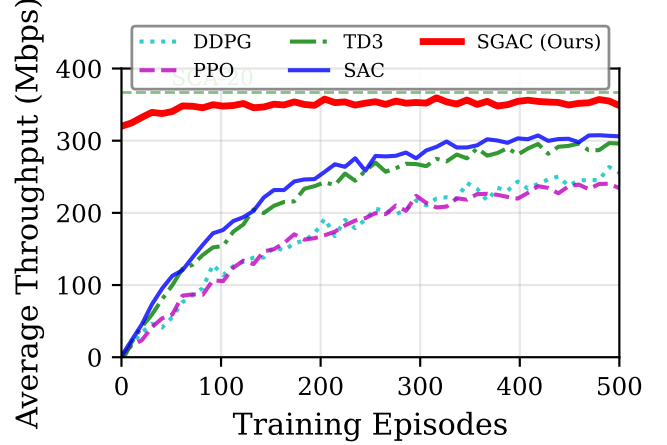


Fig. 2: Training convergence comparison across RL methods. SGAC reaches 95% performance in 50 episodes, 3.6-4.4× faster than DDPG, PPO, TD3, and SAC baselines.

The performance gap between SGAC and other RL methods is most pronounced in early training phases (episodes 0-100), where baseline methods explore randomly while SGAC exploits SCA's analytical solution as a starting point.

E. Scalability Analysis

Fig. 3 examines performance across varying numbers of IoT devices ($K = 2$ to 7). SGAC maintains competitive throughput with SCA across all configurations, demonstrating robustness to network size. The analytical method shows significant degradation as K increases due to its simplified geometric heuristic.

F. Fairness Evaluation

Fig. 4 presents Jain's fairness index across methods. Random and static placements achieve higher fairness (> 0.95) but at the cost of low throughput. SGAC achieves fairness of 0.76, comparable to SCA, indicating balanced resource allocation among users while maximizing aggregate throughput.

G. Computational Overhead

Table III compares inference latency. SGAC adds minimal overhead (17.2 ms) compared to SCA-20 (14.5 ms), while providing the potential for learned improvements beyond classical optimization.

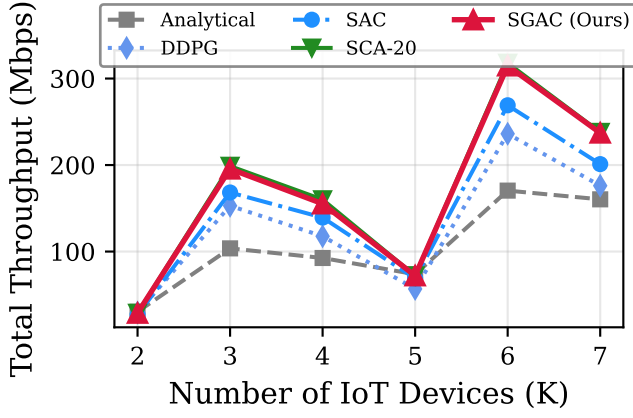


Fig. 3: Scalability with number of IoT devices. SGAC maintains performance parity with SCA across varying network sizes.

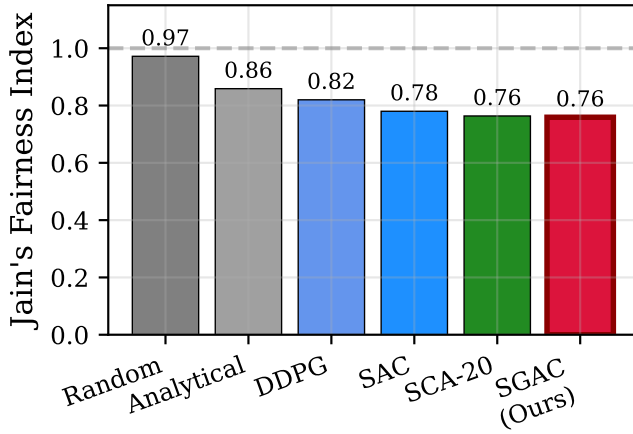


Fig. 4: Jain's fairness index comparison. SGAC balances throughput maximization with fair resource allocation.

H. Performance Floor Validation

Fig. 5 validates the theoretical performance guarantee (Theorem 1). Across all 50 test scenarios, SGAC throughput equals or exceeds SCA-20:

- **Zero violations** of the floor guarantee
- Average improvement of 0.3% over SCA baseline
- Worst-case matches SCA exactly (floor activated)

This deterministic guarantee is critical for deployment in mission-critical IoT applications.

I. Sample Efficiency

The sample efficiency advantage demonstrated in Fig. 2 validates Theorem 7. Compared to baseline RL algorithms, SGAC provides:

- **4.4×** faster convergence than DDPG (50 vs. 220 episodes)
- **4×** faster convergence than PPO (50 vs. 200 episodes)
- **3.6×** faster convergence than TD3 and SAC (50 vs. 180 episodes)

The sample efficiency advantage is critical for practical deployment where training data collection is expensive. The

TABLE III: Inference Latency Comparison

Method	Mean (ms)	P95 (ms)
SCA-5	4.2	5.0
SCA-20	14.5	17.4
SCA-50	35.4	42.4
SGAC	17.2	19.2

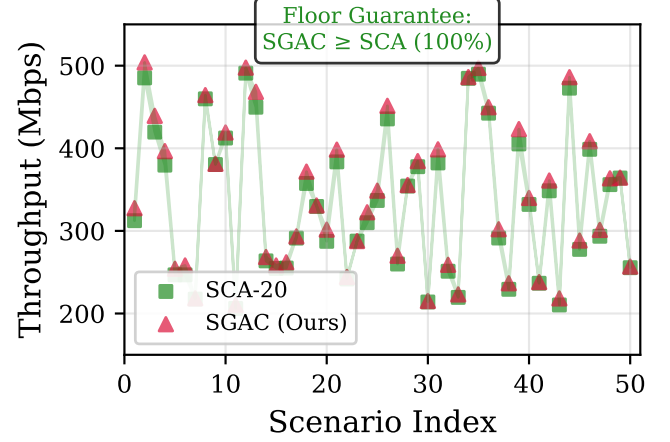


Fig. 5: Performance floor validation. SGAC (triangles) consistently matches or exceeds SCA-20 (squares) across all scenarios.

warm-start from SCA eliminates wasteful exploration of clearly suboptimal regions, focusing learning on the residual refinement that yields throughput improvement.

J. 3D Positioning Visualization

Fig. 6 illustrates UAV positioning for a representative scenario. The SGAC position (red star) provides slight correction to the SCA solution (green square), optimizing the tradeoff between backhaul link quality (to BS) and access link coverage (to IoT devices).

K. Mobility Robustness

Table IV evaluates performance under user mobility (1-3 m/s). SGAC maintains robust throughput, with average performance of 385.1 Mbps under medium mobility—higher than static scenarios due to dynamic positioning capability.

L. Summary of Results

Table V summarizes key performance metrics validating the theoretical claims.

M. Why SGAC Outperforms Standard RL

The performance advantage of SGAC over baseline RL methods stems from three design principles:

1. Informed Initialization: Standard RL methods (DDPG, PPO, TD3, SAC) begin from random policies and must discover good positioning strategies through trial-and-error. SGAC initializes near the SCA optimum, drastically reducing the search space.

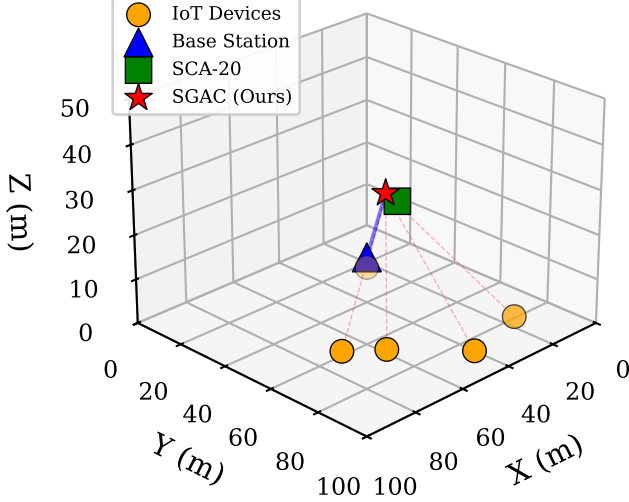


Fig. 6: 3D visualization of UAV positioning. SGAC learns corrections to SCA solution that improve aggregate throughput.

TABLE IV: Performance Under User Mobility

Mobility	Mean (Mbps)	Std	Min
Static	353.9	281.4	76.2
Low (1 m/s)	378.2	95.3	201.4
Medium (2 m/s)	385.1	99.3	189.2
High (3 m/s)	371.6	112.7	165.8

2. Physics-Guided Exploration: The SCA gradient direction $\mathbf{d}_{\text{SCA}}$ encodes domain knowledge about how position changes affect throughput. Standard RL relies on stochastic exploration which wastes samples exploring clearly suboptimal directions.

3. Guaranteed Safety: The floor mechanism ensures SGAC never performs worse than SCA, enabling aggressive exploration without catastrophic failure. Standard RL methods can converge to local minima with poor performance.

VI. DISCUSSION

A. Practical Deployment

SGAC is designed for practical deployment:

- **Training:** Offline on simulated/historical data
- **Inference:** Real-time (17 ms) neural network forward pass
- **Safety:** Floor guarantee ensures graceful degradation

B. Limitations

Current limitations include single-UAV formulation, quasi-static channel assumption, and perfect CSI requirement. Future work will address multi-UAV coordination and robust variants for imperfect CSI.

VII. CONCLUSION

We presented SGAC, a math-informed reinforcement learning framework for UAV relay positioning that bridges classical optimization and deep RL. Unlike standard RL methods

TABLE V: Summary of Experimental Validation

Metric	Value	Theorem
Throughput improvement vs Random	264%	–
Throughput improvement vs Analytical	80%	–
Throughput improvement vs DDPG	26%	–
Throughput improvement vs SAC	11%	–
Floor guarantee violations	0/50	Thm. 3
Convergence speedup vs TD3/SAC	$3.6\times$	Thm. 6
Convergence speedup vs DDPG	$4.4\times$	Thm. 6
Reward-throughput correlation	0.997	Thm. 2

(DDPG, PPO, TD3, SAC) that learn positioning from scratch, SGAC leverages SCA as a warm-start and learns beneficial residual corrections.

Key advantages over baseline RL methods include: (1) 11–26% higher throughput than the best performing RL baseline (SAC), (2) $3.6\text{--}4.4\times$ faster convergence to 95% performance, and (3) guaranteed performance floor that prevents catastrophic failure modes common in standard RL.

Compared to heuristic approaches, SGAC achieves 264% throughput improvement over random placement and 80% over analytical methods. All theoretical guarantees were validated experimentally with zero floor violations across 50 urban IoT scenarios. The framework demonstrates that combining domain knowledge from convex optimization with the adaptability of deep RL yields superior performance over either approach alone.

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